of the accelerated observers is just as correct, that is, just as valid, as that of the freely falling observers: What the freely falling observers see as virtual pairs converted into real particles by tidal gravity, followed by evaporation of one of the real particles, the accelerated observers see simply as the evaporation of one of the particles that was always real and always populated the black hole's atmosphere. Both viewpoints are correct; they are the same physical situation, seen from two different reference frames.

Figure 12.3c shows the motion of a few of the particles in a black-hole atmosphere, from the accelerated viewpoint. The particles appear to be emitted by the horizon; most fly upward a short distance and are then pulled back down to the horizon by the hole's strong gravity, but a few manage to escape the hole's grip. The escaping particles are the same ones as the infalling observers see materialize from virtual pairs (Figure 12.3a). They are Hawking's evaporating particles.

From the accelerated viewpoint, the horizon behaves like a high-temperature, membrane-like surface; it is the membrane of the "membrane paradigm" described in Chapter 11. Just as the Sun's hot surface emits particles (for example, the photons that make daylight on Earth), so the horizon's hot membrane emits particles: the particles that make up the hole's atmosphere, and the few that evaporate. The gravitational redshift reduces the particles' energy as they fly upward from the membrane, so although the membrane itself is extremely hot, the evaporating radiation is much cooler.

The accelerated viewpoint not only explains the sense in which a black hole is hot, it also accounts for the hole's enormous randomness. The following thought experiment (invented by me and my postdoc, Wojciech Zurek) explains how.

Throw into a black hole's atmosphere a small amount of material containing some small amount of energy (or, equivalently, mass), angular momentum (spin), and electric charge. From the atmosphere this material will continue on down through the horizon and into the hole. Once the material has entered the hole, it is impossible by examining the hole from outside to learn the nature of the injected material (whether it consisted of matter or of antimatter, of photons and heavy atoms, or of electrons and positrons), and it is impossible to learn just where the material was injected.